

Article

Interruption Cost Estimation for Value-Based Reliability Investment in Emerging Smart Grid Resources

Shandesh Bhattarai and Rajesh Karki * 

Department of Electrical and Computer Engineering, The University of Saskatchewan,
Saskatoon, SK S7N 5A9, Canada; bhattarai.shandesh@usask.ca

* Correspondence: rajesh.karki@usask.ca

Abstract: Growing uncertainty in supply and demand in power systems causes significant challenges in maintaining supply reliability at affordable costs. Power grids are expected to undergo substantial transformations to address these challenges with upgrades and integration of emerging smart technologies that require significant investment costs. A value-based reliability assessment of these grid technologies is necessary to justify the worth of these investments. A key parameter required in such an assessment is the cost of power interruptions originating from transmission system failures. The interruption cost data available in published reports and past surveys relate to generation inadequacy since generation facilities comprise the most capital-intensive investment of an electric utility. Customer interruptions due to a lack of generation mainly occur due to generation failures during the peak demand period, whereas interruptions due to transmission component failures can occur at other periods with specific probabilities. This paper presents a methodology to estimate the cost of outages originating from transmission asset failures, which proposes a sector period model for each customer sector to obtain associated demand-normalized interruption costs. The proposed method can also be used to decide investment in grid resiliency enhancement against extreme weather that mainly impacts the grid network facilities.

Keywords: interruption cost; reliability; resiliency; value-based investment; transmission assets



Citation: Bhattarai, S.; Karki, R. Interruption Cost Estimation for Value-Based Reliability Investment in Emerging Smart Grid Resources. *Appl. Sci.* **2024**, *14*, 8651. <https://doi.org/10.3390/app14198651>

Academic Editor: Frede Blaabjerg

Received: 13 August 2024

Revised: 23 September 2024

Accepted: 24 September 2024

Published: 25 September 2024



Copyright: © 2024 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

Modern electric power grids are undergoing substantial changes to address upcoming environmental compliance challenges while maintaining supply reliability. The replacement of fossil-fired firm generation by intermittent and uncertain renewable energy sources and the electric transformation of the automotive industry will cause increasing difficulty in balancing the electric supply and demand in future power systems. There is ongoing research and development in new and smart technologies in energy storage, power electronics, digital and communication devices that can increase the capability of power systems to continuously meet consumer demands. Long-term system planning requires the optimal selection and allocation of these technologies in the bulk power grid and the necessary network upgrades to comply with existing and future environmental regulations and targets. System planners require appropriate methodologies and assessment tools to determine optimal investment in these bulk power system facilities to maintain acceptable reliability at affordable costs. The value-based reliability investment (VBRI) method analyzes the cost and worth of reliability investment in order to justify the investment in new resources, facilities, and/or system upgrades. This method can be used with necessary modifications or extensions to address system-specific investment decision problems.

The VBRI approach requires a comparative study of two aspects of system adequacy and economics, and the resulting cost/worth analysis [1] is performed to estimate the expected investment benefits in decision-making. The adequacy cost is the investment cost needed to achieve a certain level of reliability, whereas adequacy worth is the benefit

derived by the utility, consumer, and society in the form of reduced power outages or interruption costs. It quantifies the monetary losses incurred by electricity customers due to the unreliability of the power system. An adequacy assessment is not usually conducted on the entire power system but is performed at the various sub-systems or hierarchical levels (HL) as shown in Figure 1. HL-I assessment considers only the generation system and its ability to meet the overall system demand. HL-II or bulk system adequacy refers to the existence of adequate generation and transmission facilities to meet the bulk load requirements distributed throughout the grid network. HL-I VBRI assessment helps make the investment decision on appropriate and adequate generation resources to minimize the overall societal cost, i.e., the sum of the investment cost and the outage cost as shown in Figure 2 [2]. Similarly, HL-II VBRI studies can be conducted to make investment decisions on transmission facilities while considering both the generation and the transmission systems. VBRI assessment techniques for HL-I and HL-II are illustrated in [2].

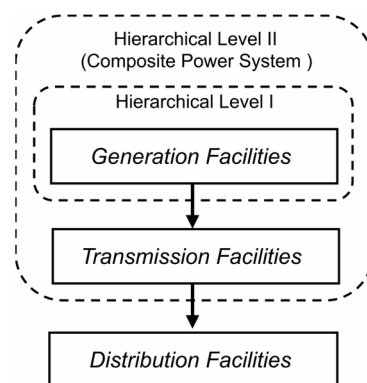


Figure 1. Composite Power System.

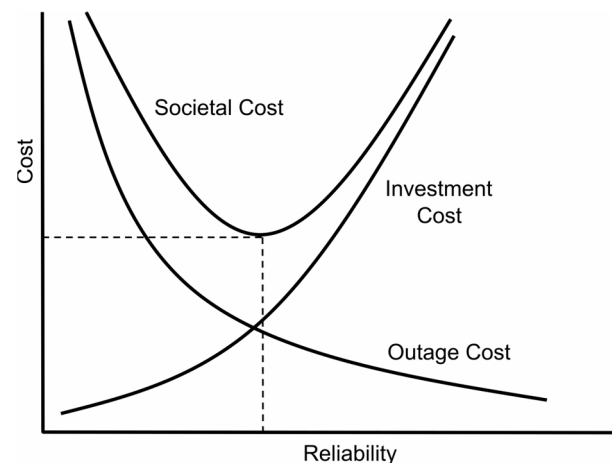


Figure 2. Value-based reliability investment approach.

Significant work [3–12] has been performed on VBRI methods and their applications at the HL-I level. Generation facilities comprise the most capital-intensive investment of an electric utility in expansion planning. The interruption cost estimation in a value-based reliability assessment is therefore primarily focused on generation inadequacy. The interruption cost can be estimated using various techniques [13], such as analytical methods, case studies of blackouts, and customer surveys. Among these methods, customer surveys [14–20] provide the most reliable information related to the usage of electricity and the impact of power outages on the different categories of electricity customers. The customer survey is designed primarily to obtain relevant information to assess the monetary losses associated with various frequencies and duration of power outages at different times of the day and year. The customers are categorized into different sectors, such as residential,

commercial, industrial, etc. The survey data are used to estimate the interruption cost in CAD/interruption for each customer sector for a range of interruption durations. This value is then normalized by the sector load demand to obtain the demand-normalized cost (DNC) in CAD/kW. The DNCs of a customer sector calculated for the selected outage durations form a customer damage function (CDF) [21] for that customer sector. The CDFs of all the different customer sectors are aggregated to obtain the composite customer damage function (CCDF) [1], which can then be used in alternate HLI expansion plans to determine the corresponding outage costs. The cost of interruption during the system peak becomes the deciding factor in determining the capacity reserve needed above the peak demand [22]. The interruption costs are therefore normalized by the peak demand in VBRI assessment at the HL-I level. The CDFs thus obtained are not suitable for VBRI assessment of transmission or distribution systems and facilities.

Transmission planning usually follows HL-I expansion planning and makes use of deterministic reliability methods, such as the North American Reliability Corporation's N-1 planning criteria [23]. The rapid growth in renewable/intermittent generation and the expected growth of electric vehicles will significantly increase uncertainties in line loadings and net demands at the bulk load points. There are considerable efforts and activities [24] among electric utilities and research institutions to adopt probabilistic reliability methods that recognize these uncertainties in HL-II system planning and decision-making. An appropriate VBRI must be conducted to determine HL-II reliability criteria that yield minimum societal costs. The increased variability and uncertainty in the bulk electric systems require additional support resources and emerging smart technologies to operate the system in a reliable manner. For example, various types of energy storage systems, voltage support devices, and smart-grid technologies to acquire and process relevant system data for operational management of power supply and demand will be expected to be deployed throughout the network. A VBRI assessment at the HL-II level will be necessary to justify the cost and worth of deploying these new technologies into the bulk system network. Also, the increase in frequency of extreme weather events in past decades due to climate change [25] as well as other extreme events like cyber-physical attacks and seismic events have impacted the grid facilities due to their exposure to these events [26] causing catastrophic losses [27,28]. The cost and worth of hardening the grid infrastructure against such adverse events requires an appropriate HL-II VBRI assessment.

There has been limited work on VBRI at the transmission system level. A VBRI approach is illustrated in [29,30] to make expansion planning decisions on a composite generation and transmission system. Ref. [31] presents a methodology to determine the optimum probabilistic reliability criterion for grid expansion planning to meet a specified level of reliability at the lowest cost. A VBRI approach is introduced in [32] for determining optimal reliability criteria considering the load bus reserve rate in a long-term load forecast for optimal transmission systems expansion planning with minimal costs for society. An application of a value-based reliability model is presented in [33] for transmission capital projects in a real transmission system with feasible transmission reinforcement alternatives that provide reliable electricity at the lowest possible rate. Ref. [34] proposes a VBRI approach to determine line upgrades to alleviate transmission congestion and maintain the reliability of a bulk system integrated with large-scale wind power. Ref. [35] describes the reliability worth scheme for multiple dispatch scenarios using binary disjunctive techniques and screening strategies. A multi-stage planning of transmission system to minimize overall societal cost is proposed in [36] based on the metaheuristic Ant Colony Optimization. These studies, however, use a customer outage cost model that was prepared for HL-I, which is not appropriate for outage cost estimation associated with transmission-related outages. Customer interruptions due to transmission-related outages are not confined to the peak load period as in HL-I but can occur at other times with a relatively high probability depending on weather patterns and loading conditions of individual lines at different times of the day and season. It should further be noted that the cost of a power outage can vary widely at different times of the day and season. For example, the power interruption cost to

a commercial customer during business hours is much higher than that outside of business hours. Also, the cost of a power interruption to a residential customer is generally low during a mild spring evening when compared to a hot summer afternoon. The outage cost model for HL-II can therefore be very different from the model obtained for HL-I. Ref. [37] provides estimates of customer damage functions that can be applied to calculate interruption costs per event by season, time of day, day of week for various electricity customers in the United States. However, in [37], the outage cost was normalized by the average demand and unable to obtain a demand-normalized cost of power interruptions for a range of possible load demands in the study period. Ref. [38] proposes a VBRI methodology for transmission expansion planning by using customer-surveyed outage cost data and historical annual load profiles divided into the summer period, winter period, and off-peak period. This study lacks the consideration of diurnal load variation, as power outage costs at different times of the day can vary considerably as mentioned above. It is therefore important to recognize the impact of transmission system-originated failures on customer interruptions, to estimate the frequency and duration of their occurrence at different times of the day, season, and year, and to develop a VBRI methodology to incorporate them in a bulk system reliability model.

This paper presents a methodology to develop a composite customer damage function (CCDF) for VBRI assessment of a bulk electric power system that considers customer interruptions arising from transmission system-originated failures. The proposed method incorporates the probability of a power outage occurrence at different times of the day and year and the impact of the time of occurrence and duration of the outage on the customer interruption cost.

2. Estimation of Interruption Costs Arising from Transmission System Originated Failures

This section describes the methodology to estimate the cost of interruptions incurred at the bulk load points due to forced outages of components of a bulk power system. As earlier mentioned, the customer interruption costs obtained from previous works [29–36] are mainly applicable to the determination of optimal investment in generation capacity reserves in a power system. It should be noted that power utilities decide investment in generation based on the capacity reserve requirement, or a generation adequacy criterion, such as NERC's loss of load expectation (LOLE) criterion. The peak demand dictates the capacity reserve and is most sensitive to the LOLE index. Value-based investment in generation resources, therefore, utilizes customer outage costs during the peak demand. Existing customer damage functions available to the power industry are functions of the peak-demand-normalized costs with outage durations. A new approach is proposed in this paper that considers the fact that power outages arising from the failures of bulk system components, such as transmission lines or devices connected to the bulk power grid, can occur at any time of the day or year with a probability that can be estimated from past performance data. The new approach also recognizes the fact that the energy consumption patterns vary significantly depending on the type of electricity customer and so do the interruption costs. For example, a residential customer load is characterized by a specific diurnal and seasonal variation pattern, whereas an industrial customer load is more or less constant throughout the year. The bulk system customers are divided into different sectors, such as industrial, residential, commercial, farm, etc. The proposed methodology consists of (1) developing a sector periodic demand model, and (2) developing an integrated interruption cost model, in order to obtain the cost of interruption originating from HL-II components.

2.1. Sector Periodic Demand Model

The interruption costs in VBRI assessment are expressed as demand-normalized costs (DNC), and therefore, the proposed method utilizes a periodic demand model to estimate the demand-normalized interruption costs for the bulk system customer sectors. Figure 3

shows the steps involved to obtain the sector periodic demand (SPD) model for a bulk system. The proposed model is designated as the SPD model in this paper.

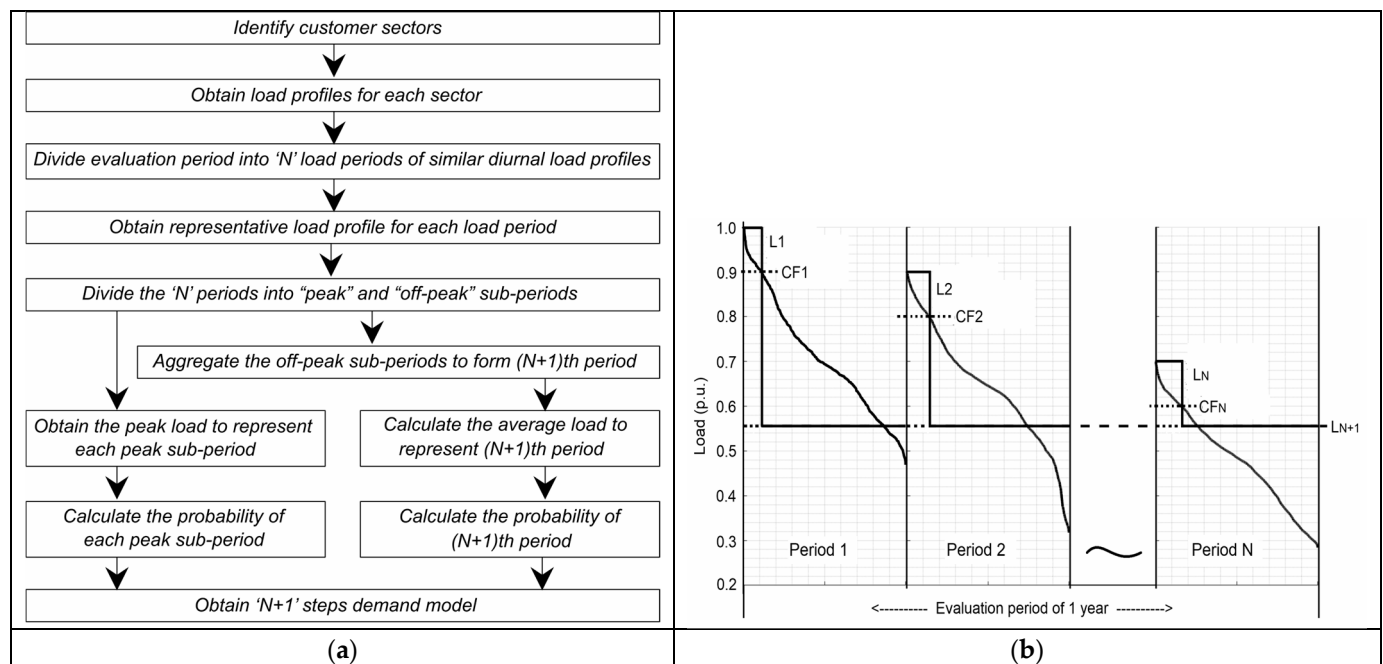


Figure 3. Development of a sector periodic demand model. (a) Methodology steps, (b) Sub-division of the evaluation period.

The first step of the methodology is to identify the major customer sectors in a bulk system. Electricity consumers are classified into various customer sectors depending on their end-use applications. Typically, customers within a specific sector share similar needs for electricity and are affected similarly by power outages. Residential, commercial, and industrial sectors are a significant part of a bulk system load. However, specific customer sectors, such as farm or oil sectors, may exist in bulk systems based on the region they operate.

The next step is to obtain the historical load data for each customer sector identified in the previous step. The acquisition of customer sector load data is a part of utility load research [39,40], which is used to analyze cost-of-service, load forecasting, load management technologies, and other system planning-related activities. In recent years, customer sector load data can be obtained from the advanced metering infrastructure (AMI) in various time intervals. As the proposed methodology is aimed at emerging power systems transitioning to meet low-emission environmental targets, the intermittent renewable generation and electric vehicle loads are incorporated in the load profile of the customer sector that owns them. The proposed methodology in this paper utilizes the hourly interval load data expressed in kilowatts (kW) as given by (1).

$$D_i = \{H_{i,h} \mid i = 1, 2, \dots, d \text{ and } h = 1, 2, \dots, 24\} \quad (1)$$

where the daily load profile D is a set of 24-hourly loads H , and d is the total number of days of historical data used in the analysis.

Adequacy assessment in system planning utilizes annual reliability indices, and therefore, the evaluation period is typically one year. The sector load data obtained are then grouped into N periods such that the diurnal loads with identical profiles are grouped into one period as described in (2).

$$\{D_{i,p=1,2,\dots,N}\} \in \{D_i\} \mid \mu(D_{i=j}) \approx \mu(D_{i \neq j}) \text{ and } med(D_{i=j}) \approx med(D_{i \neq j}) \text{ for } j = 1, 2, \dots, d \quad (2)$$

where functions $\mu()$ and $med()$ evaluate the mean and median of D_i .

The proposed methodology then determines a representative load profile for each of the N periods. It should be noted that increasing the number of periods will improve the accuracy of the SPD model. This will, however, reduce the accuracy of the survey results due to the increased complexity in outage cost estimation by the customers at increasing time resolutions. A trade-off is therefore required. The bulk system reliability indices are highly sensitive to the peak load since the system is vulnerable to customer outages due to generation failures during minimum reserve conditions and transmission failures during line congestion. The load profiles in a period are therefore represented by one load profile RD_p with the highest peak load as shown in (3). The representative load profile is then sorted in descending order.

$$RD_p = D_{i=k,p} | \max(\{H_{k,h}\}) > \max(\{H_{i \neq k,h}\}) \quad (3)$$

The contribution of transmission system failures to customer outages is highly dependent on the network structure and configuration, and such outages can occur during off-peak hours as well. The next step of the proposed methodology is therefore to subdivide each of the N periods into peak and off-peak sub-periods using a cut-off factor (CF_p) as shown in Figure 3b and calculated in (4) for each period p .

$$CF_p = 0.9 \times \max(\{H_h\}) | H_h \in RD_p \quad (4)$$

The peak load, L_p , is next determined as the representative load for the peak sub-period p as shown in (5). The probability of occurrence of the representative load L_p is given by (6).

$$L_p = \max(\{H_{i,h}\}) | H_{i,h} \in RD_p \quad (5)$$

$$Prob_p = \frac{|\{H_{i,h} | H_{i,h} \in \{D_{i,p}\}, H_{i,h} \geq CF_p\}|}{d \times 24} \quad (6)$$

The load profiles $D_{i,N+1}$ in the $(N + 1)$ th demand period are next obtained by aggregating load data from the N off-peak sub-periods as given by (7). The outages during off-peak sub-periods are due to random failures of bulk system components. The average load given by (8) is therefore taken as the representative load, L_{N+1} in the $(N + 1)$ th period. The probability of occurrence of the representative load L_{N+1} is given by (9).

$$D_{i,p=N+1} = \bigcup_{p=1}^N \{H_{i,h} | H_{i,h} \in \{D_{i,p=1,2,\dots,N}\}, H_{i,h} < CF_p\} \quad (7)$$

$$L_{p=N+1} = \frac{1}{|\{H_{i,h} | H_{i,h} \in \{D_{i,p=N+1}\}\}|} \sum_{H_{i,h} \in \{D_{i,p=N+1}\}} H_{i,h} \quad (8)$$

$$Prob_{p=N+1} = \frac{|\{H_{i,h} | H_{i,h} \in \{D_{i,p=N+1}\}\}|}{d \times 24} \quad (9)$$

Finally, the ' $N + 1$ ' step discrete probability distribution is obtained from the representative load level and its associated probability derived from (5), (6), (8), and (9) from the SPD model as shown in (10).

$$\{(L_p, Prob_p) | p = 1, 2, \dots, N + 1\} \quad (10)$$

The proposed methodology is developed with the objective of its application in emerging power systems transitioning to meet low-emission environmental targets. The intermittent renewable generation and electric vehicles are incorporated in the load profile of the customer sector that owns them. Since the proposed methodology incorporates an SPD model that includes clusters of identical diurnal load profiles distributed across the annual evaluation period, it can be applied to any type of power system. The number of

sub-periods would differ based on the power system net load variability in low-carbon power systems, whereas the SPD model of a conventional power system can be divided into seasonal sub-periods, as the load profiles in each season are more or less similar.

2.2. Integrated Interruption Cost Model

This section describes the methodology to develop the interruption cost model for each customer sector that can be integrated with the SPD model described in Section 2.1. The interruption cost incurred to a customer sector due to bulk component failure depends on the magnitude of the load curtailed and the duration of the interruption. As the magnitude of the load varies with time with a characteristic specific to a customer sector, it will be necessary to determine the interruption cost as a function of time of day, time of year, and the interruption duration. This information can be extracted from a customer survey [18] of all the customer sectors within the power system. The outage cost estimation from the customer survey assumes that the customers are in the best position to assess the cost of their outage that may occur at different times and scenarios. An inaccurate response from a customer can deviate from the survey results, and therefore, careful screening is required to identify and filter out outliers from the survey sample. It should be noted that the outage cost derived is the best possible estimate, as intangible and indirect costs associated with emotional and other social values are very difficult to quantify. The statistical estimation method [14–16] is applied to obtain the interruption cost IC_p at each of the 'N + 1' periods p of the SPD model for a range of outage durations as shown in (11).

$$IC_p = \{ic_{p,t} \mid t = 1, 2, \dots, T\} \quad (11)$$

where, $ic_{p,t}$ is the interruption cost in CAD/interruption for an outage of duration t during the load period p . The outage duration, $t = 1, 2, \dots, T$, representing 1 min, 20 min, 1 h, 2 h, 4 h, 8 h, 24 h [14–16,41,42] are usually considered in these studies.

The interruption cost is then normalized by the representative load for each period of the SPD model. The demand-normalized cost $DNC_{p,t}$ in CAD/kW is obtained from (12),

$$DNC_{p,t} = \{ic_{p,t}/L_p\}_{\text{for } p=1 \text{ to } N+1 \text{ and } t=1 \text{ to } T} \quad (12)$$

The expected demand-normalized cost $EDNC_t$ for an outage duration t is next obtained from (13) by taking the weighted sum of all the DNC calculated for all the N + 1 demand periods of the SPD model.

$$EDNC_t = \sum_{p=1}^{N+1} (\text{Prob}_{p,t} \times DNC_{p,t}) \quad (13)$$

The customer damage function (CDF) is the expected demand-normalized cost as a function of the interruption duration. The CDF for a sector, SCDF, is given by (14):

$$SCDF = \{EDNC_1, EDNC_2, \dots, EDNC_t, \dots, EDNC_T\} \quad (14)$$

The composite customer damage function (CCDF) at a bulk load point is calculated by weighting the SCDF from (14) by load compositions (peak load and energy consumption percentages) of customer sectors as given by (15).

$$CCDF = \left\{ \sum_{m=1}^M (EDNC_{m,t} \times wf_{m,t}) \right\}_{\text{for } t=1 \text{ to } T} \quad (15)$$

where $m = 1, 2, \dots, M$ denotes the participating customer sectors, and $wf_{m,t}$ is the weighting factor for a customer sector at the load point such that [41]

$$wf_{m,t} = \begin{cases} L_{peak,m}/L_{peak} & \text{for } t < 1 \text{ hour} \\ E_m/E & \text{for } t \geq 1 \text{ hour} \end{cases}$$

where $L_{peak,m}$ and E_m are the annual peak load and energy consumption of Sector m , respectively, and L_{peak} and E are the annual peak load and energy consumption of the overall system.

Finally, the monetary equivalence of power loss is expressed as the expected cost of customer interruptions (ECOST) as given by (16).

$$ECOST_k = \sum_{i=1}^{NC} L_{ki} \times F_i \times C_i(D_i) \frac{MW}{kW} \$/\text{yr} \quad (16)$$

where L_{ki} , F_i , and $C_i(D_i)$ represent the magnitude (MW), frequency (occ/year), and CCDF (CAD/kW) of an outage of duration D_i from contingency i at load point k , respectively. Also, NC represents the total number of outages leading to the power interruptions at load point k .

3. Illustration of the Methodology

The section describes the illustration of the methodology using the load data from the Saskatchewan province in Canada and the outage cost data from the survey [43] conducted by the University of Saskatchewan for generation adequacy assessment. Sections 3.1 and 3.2, respectively, illustrate the development of the SPD model and the integrated interruption cost model. Although the proposed methodology is aimed at low-carbon bulk power systems, it can be applied to all types of power systems. This is because the SPD model includes clusters of identical diurnal load profiles distributed across the annual evaluation period. The number of periods in the SPD model will differ based on the net load variability. Since conventional power systems have approximately similar load profiles in each season, the SPD model can be divided into four seasonal periods. The application of the proposed methodology is illustrated for an annual evaluation period. System planning typically considers a planning horizon of 10 or more years. The annual evaluation is therefore repeated over the planning horizon using forecast load data provided by the power utility in a practical application of the methodology.

3.1. Obtaining the SPD Model

The initial step of the methodology requires identifying the major electricity customer categories based on their end-use applications. Residential, industrial, commercial, and farm customers were identified as the major categories in the Province of Saskatchewan. Historical hourly variations of the load data for each identified customer sector are shown in Figure 4 per unit of the respective peak values. The annual load profile for each sector was then divided into periods having similar diurnal load profiles. Figure 4 shows that the residential and farm sectors were divided into four seasonal periods since the diurnal load profiles for these customer sectors had reasonable similarity as assessed from (2). The diurnal load profiles of the industrial and commercial sectors were found to be similar throughout the year.

The next step is to observe the diurnal load variation in each period and further subdivide it into peak and off-peak periods. Figure 5 shows the subdivision using a cut-off factor of 0.9 per unit of the peak load in that period. The figure shows that the commercial sector is divided into two periods that are represented by the peak and average loads in the respective periods calculated using (5) and (8), respectively, and the probabilities of encountering the two periods are calculated using (6) and (9), respectively. Figure 5 also shows that diurnal variation in the industrial load is insignificant, and, therefore, was not subdivided.

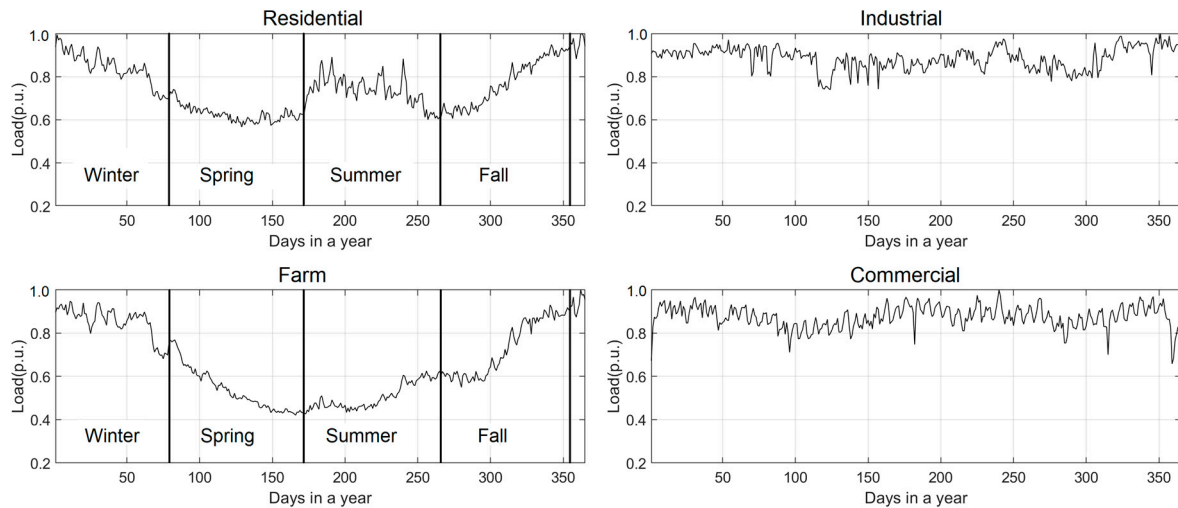


Figure 4. Annual load profiles for the identified customer sectors.

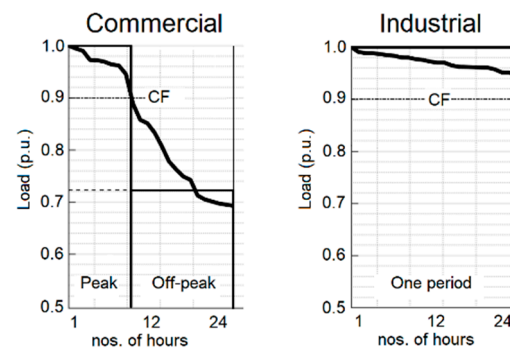


Figure 5. Diurnal load variation of commercial and industrial sectors.

Figure 6 shows the diurnal load profiles for the residential and farm sectors in the four seasonal periods. The seasonal periods for the residential and farm sectors are similarly further subdivided into peak and off-peak periods as shown in the figure. The representative load and its probability are similarly calculated using (5) and (6), respectively.

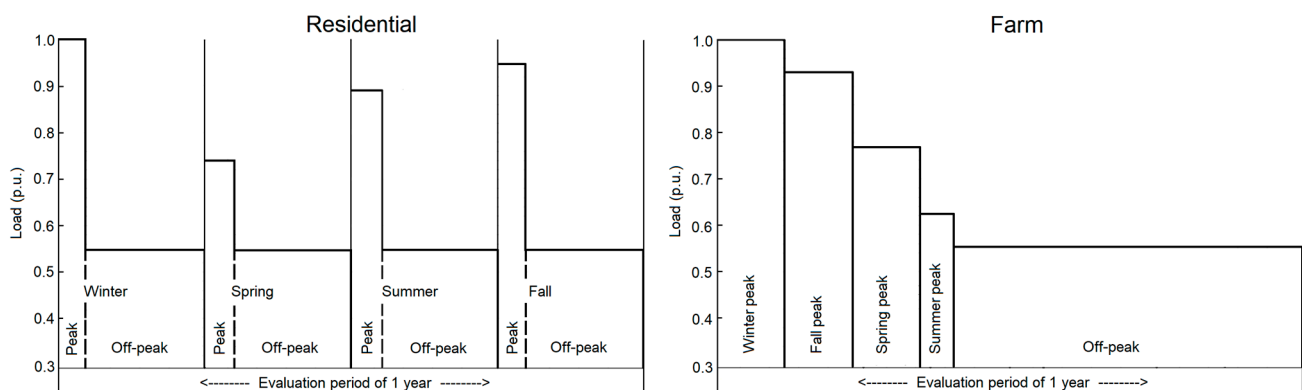


Figure 6. Periodic load models for the residential and farm sectors.

The off-load sub-periods in the residential sector are grouped together into one period, and the representative load and its probability are calculated using (8) and (9), respectively. The load and corresponding probabilities for the different periods of the farm sector are obtained similarly. The SPD model thus obtained is shown in Table 1.

Table 1. SPD Model.

Period	Load (p.u. of Sector Peak)	Probability
Residential		
Winter peak	1.0000	0.009247
Spring peak	0.7410	0.003767
Summer peak	0.8913	0.003196
Fall peak	0.9489	0.010300
Off-peak	0.5470	0.973500
Farm		
Winter peak	1.0000	0.011758
Spring peak	0.7685	0.006279
Summer peak	0.6254	0.006507
Fall peak	0.9305	0.019064
Off-peak	0.5538	0.956393
Commercial		
Peak	1.0000	0.068493
Off-peak	0.7227	0.931507
Industrial		
One period	1.0000	1.000000

3.2. Obtaining the Integrated Interruption Cost Model

The data obtained from the outage cost survey [43] conducted in the year 2017 for Saskatchewan electricity customers are used to illustrate the interruption cost model. The cost per interruption data obtained for six different outage durations as shown in Table 2 are used in this study. The interruption cost data from the survey are converted to the 2023 price equivalence using the inflation calculator in [44]. It should be noted that the interruption cost data for all the required time periods and outage durations were not directly available from the 2017 survey since it was targeted to obtain data for generation planning. The data in Table 2 therefore pertain to the winter peak period, which can be expressed as shown in (11). An approximate cost estimation method as described in this section was followed to indirectly obtain the relevant interruption costs incurred at other periods of the years.

Table 2. Interruption cost (CAD/Int) data from the 2017 survey (in 2023 CAD).

Sector	1 min	20 min	1 h	4 h	8 h	24 h
Residential	0.00	0.54	4.06	45.52	92.78	352.07
Farm	1.83	8.06	13.98	24.32	26.83	52.37
Commercial	29.69	87.81	237.21	1194.95	1920.18	2312.68
Industrial	252,912.00	417,959.00	822,855.00	2,719,105.00	5,239,916.00	15,431,217.00

The next step is to normalize the sector interruption cost by the load demand at each time period of the SPD model to obtain the DNC using (12). Table 3 shows the annual peak loads of the different customer sectors considered in the study.

Table 3. Sector annual peak load, KW.

Sector	Residential	Farm	Commercial	Industrial
Sector peak (kW)	2.0461	2.7000	8.1000	99,780.0000

The SPD model of the industrial sector consists of only one period as shown in Table 1. The DNC of the industrial sector obtained using (12) is shown in Table 4.

Table 4. Industrial DNC, CAD/KW.

Sector	1 min	20 min	1 h	4 h	8 h	24 h
One period	2.53	4.19	8.25	27.25	52.51	154.65

Residential, farm, and commercial sectors have more than one load period in the SPD model. The interruption cost data for periods other than the winter peak period were not available from the survey [43] results. The approximate cost estimation method to obtain the interruption costs for the other periods in the SPD model is illustrated for the residential sector. A similar approach was used for the other remaining sectors.

The 2017 survey also obtained data on preparatory action costs of five alternative electric supplies shown in Table 5 and the six-point Likert Scale shown in Table 6 to reflect the undesirability of power outages in different outage scenarios. The 4 h outage duration undesirability Likert Scale for weekly and monthly outage frequency during different seasonal periods obtained from the survey is shown in Table 7. These data are used in the approximate method to estimate the interruption costs during other periods of the SPD model. It should be noted that the residential sector has five periods in the SPD model as shown in Table 1.

Table 5. Preparatory action costs from the survey.

Actions	No Action	Provide Minimal Lighting	Provide Full Lighting and LED TVs	Lighting, TVs and Small Appliances (Fan)	Full Household Load
Cost (CAD/h.)	0.0	1.0	6.0	12.0	25.0

Table 6. Six-point Likert Scale from the survey.

Undesirability	None	Low	Moderate Low	Moderate High	High	Extremely Undesirable
Scale	1	2	3	4	5	6

Table 7. Likert Scale (4 h duration) for various scenarios.

Likert Scale (4 h)	1	2	3	4	5	6	Cost, CAD/int
Winter peak (weekly)	4	3	5	14	35	162	82.28
Winter peak (monthly)	11	22	39	41	53	57	45.97
Winter off-peak (weekly)	6	12	15	31	45	113	65.82
Spring peak (weekly)	14	12	35	50	51	61	47.70
Summer peak (weekly)	10	30	39	51	37	56	43.30
Fall peak (weekly)	10	13	24	58	55	62	48.92

The six-point Linkert scale is reduced to a five-point scale by merging Scale 3 (moderate-low) and Scale 4 (moderate-high) together. This is performed to match the five-point scale to the five preparatory action costs. The interruption costs for the different seasonal periods

listed in the first column of Table 7 are estimated by the weighted average of the preparatory actions cost per hour by the undesirability rating of the customer responses. The last column of Table 7 shows the interruption costs estimated using the approximate method. Equation (17) shows the calculation of the interruption cost of a 4 h outage occurring monthly during the winter peak period. It can be seen that this value is close to the 4 h interruption cost for the residential sector shown in Table 2 and is therefore used as the interruption data for the winter peak period of the proposed model.

$$\text{Cost (\$/int)} = \frac{4\text{h}}{1\text{h}} \times \frac{(11 \times 0 + 22 \times 1 + (39 + 41) \times 6 + 53 \times 12 + 57 \times 25)}{11 + 22 + (39 + 41) + 53 + 57} = 45.97 \quad (17)$$

The Table 7 data shows that the monthly occurring outage cost is 56% of the weekly outage cost. This ratio can be used to estimate the 4 h outage cost data shown in Table 8 for the remaining peak periods of the proposed residential SPD model. Similarly, Table 7 shows that the winter off-peak cost is 80% of the winter peak data. The outage during the off-peak period is therefore estimated to be 80% of the peak period. Since all the off-periods are merged together in the SPD model, the average sum of the off-peak costs as calculated in (18) is used as the 4 h outage cost for the off-period in the SPD model.

$$\text{Off-peak (\$/int)} = \frac{45.97 + 26.71 + 24.25 + 27.40}{4} \times 80\% = 24.87 \quad (18)$$

Table 8. Interruption cost of 4 h outage during the peak periods of the residential sector.

Period	(2023, CAD/int)
Winter peak	45.97
Spring peak	26.71
Summer peak	24.25
Fall peak	27.40
Off-peak	24.87

The next step is to calculate the interruption cost for outage durations other than 4 h. Table 2 shows the interruption costs for outage durations of 1 min, 20 min, 1 h, 8 h, and 24 h are 0.0%, 1.2%, 8.9%, 203.8%, and 773.5%, respectively, of a 4 h duration for the residential sector. The calculated outage costs as expressed by (11) during the periods of the residential SPD model are shown in Table 9.

Table 9. SPD model interruption cost for residential sector.

Period	1 min	20 min	1 h	4 h	8 h	24 h
Winter peak	0.00	0.55	4.09	45.97	93.69	355.58
Spring peak	0.00	0.32	2.38	26.71	54.43	206.60
Summer peak	0.00	0.29	2.16	24.25	49.42	187.57
Fall peak	0.00	0.33	2.44	27.40	55.84	211.94
Off-peak	0.00	0.30	2.21	24.87	50.69	192.37

A similar approach was used to obtain the interruption costs shown in Table 10 for the farm and commercial sectors. Since the only data available from the survey for the farm and commercial sectors are the interruption cost data shown in Table 2, these cost data are taken as the base values to approximate the costs in Table 10 using a linear approximation from the residential cost data shown in Table 8.

Table 10. SPD model interruption cost for farm and commercial sectors.

Period	1 min	20 min	1 h	4 h	8 h	24 h
Farm						
Winter peak	1.83	8.06	13.98	24.32	26.83	52.37
Spring peak	1.06	4.68	8.12	14.13	15.59	30.43
Summer peak	0.97	4.26	7.38	12.84	14.17	27.65
Fall peak	1.09	4.80	8.33	14.49	15.99	31.20
Off-peak	0.99	4.36	7.56	13.16	14.52	28.34
Commercial						
Peak	29.69	87.81	237.21	1194.95	1920.18	2312.68
Off-peak	16.03	47.52	128.32	646.47	1038.81	1251.18

The interruption costs are then normalized by the sector demand during each period of the SPD model obtained by multiplying the per unit demand by the sector annual peak load in Table 3. The sector DNCs calculated using (12) for residential, farm, and commercial are presented in Table 11.

Table 11. Residential, farm, and commercial DNCs, CAD/KW.

Period	1 min	20 min	1 h	4 h	8 h	24 h
Residential						
Winter peak	0.00	0.27	2.00	22.47	45.79	173.78
Spring peak	0.00	0.21	1.57	17.62	35.90	136.26
Summer peak	0.00	0.16	1.18	13.30	27.10	102.85
Fall peak	0.00	0.17	1.26	14.11	28.76	109.16
Off-peak	0.00	0.27	1.97	22.22	45.29	171.88
Farm						
Winter peak	0.68	2.99	5.18	9.01	9.94	19.40
Spring peak	0.51	2.26	3.91	6.81	7.51	14.67
Summer peak	0.57	2.52	4.37	7.60	8.39	16.37
Fall peak	0.43	1.91	3.32	5.77	6.36	12.42
Off-peak	0.66	2.92	5.06	8.80	9.71	18.95
Commercial						
Peak	3.67	10.84	29.29	147.52	237.06	285.52
Off-peak	3.00	8.89	24.00	120.93	194.32	234.04

Next, the sector customer damage function (SCDF) is formed from the expected DNCs for each sector. The industrial sector has one load period, and therefore, the expected DNC is the same as the data in Table 4. The expected DNCs for residential, farm, and commercial sectors are obtained by (13). The SCDF as described in (14) is presented in Table 12.

The composite customer damage functions (CCDFs) for all load points are finally calculated using (15). The composite power system consists of multiple load points with unique load compositions. Therefore, the CCDFs will vary from one load point to another depending upon the composition of the customer sectors at a given load point. Figure 7 shows the 17 load points of the IEEE-RTS 24-bus system, which has an annual peak demand of 2850 MW and a load factor of 61.4%. Table 13 shows the load sector composition for two selected load points, 4 and 14, that are assumed to illustrate the method. The CCDF obtained at these load points is shown in Figure 8.

Table 12. Sector customer damage function (SCDF), CAD/KW.

Sector	1 min	20 min	1 h	4 h	8 h	24 h
Residential	0.00	0.27	1.96	22.09	45.03	170.90
Farm	0.65	2.89	5.02	8.72	9.63	18.79
Commercial	3.05	9.02	24.36	122.75	197.25	237.57
Industrial	2.53	4.19	8.25	27.25	52.51	154.65

The CCDF can be used to assess the cost of outages in grid reliability planning or in deciding grid resilience enhancement investments. For example, the southwest corner of the test system in Figure 7 lies in the high wind zone, and an option to reinforce the tower structures in the region is considered to enhance the resiliency against extreme winds. A resilience study can be conducted to estimate the duration and magnitude of load curtailment at each node and used in (16) to calculate the ECOST of the system. This analysis is conducted for two cases: one with tower reinforcement and the other without reinforcement. The reduction in total outage cost due to tower hardening is compared with the reinforcement cost to make an investment decision. Another example is a planning decision to twin a transmission line between Buses 15 and 24 to enhance grid reliability. A composite system reliability assessment of the test system can be carried out with and without considering line twinning, and the resulting duration and MW are curtailed at each load point and then used in (16) to assess the outage costs for the two cases. The decision to invest in a new transmission line should then be justified by the reduction in ECOST due to the investment.

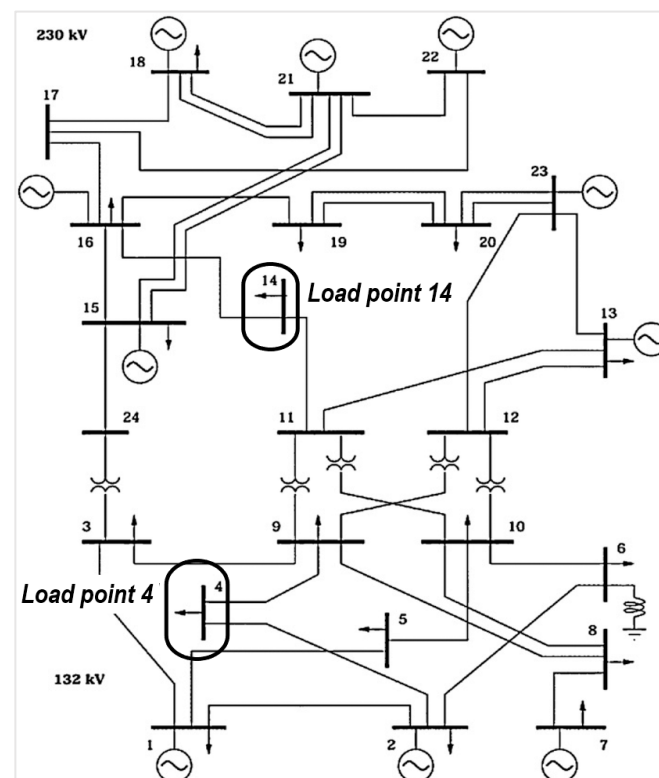
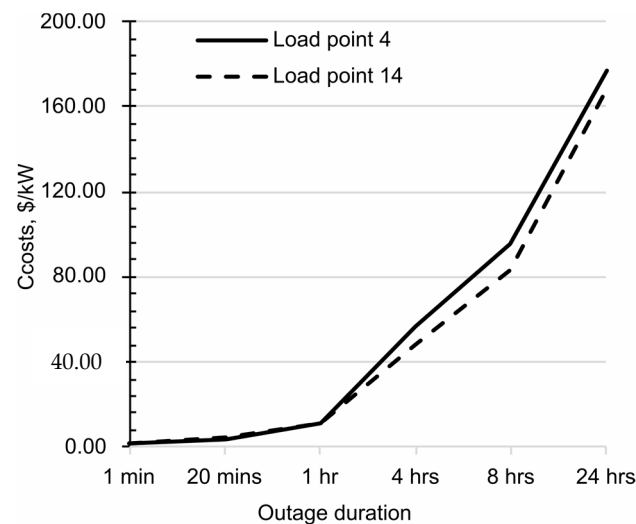
**Figure 7.** IEEE-RTS 24-bus system.

Table 13. Assumed sector load composition at two load points in IEEE-RTS 24-bus system.

Sector	Peak Demand %		Energy Consumption %	
	Load Point 4	Load Point 14	Load Point 4	Load Point 14
Residential	40%	19%	45%	14%
Farm	15%	10%	10%	6%
Commercial	30%	25%	35%	24%
Industrial	15%	46%	10%	56%

**Figure 8.** CCDF at two different load points of the IEEE-RTS system.

4. Conclusions

Substantial investments in transmission resources will be required to address the growing challenges of maintaining the reliability of power grids, stemming from the rising penetration of intermittent renewable energy sources and the growth in electric vehicles causing increased uncertainty in load profiles across the load centers. Ongoing research and development in smart technologies, such as energy storage, power electronics, and digital communication devices, provide opportunities for enhancing the capabilities of the transmission system. The appropriate investment in the selected emerging technologies should, however, enhance the grid network reliability and mitigate the cost of power outages. The methodology proposed and illustrated in this paper can be used by grid reliability planners to estimate the cost of outages originating from transmission facilities. The outage costs obtained from the proposed method can also be used to decide investment in grid resiliency enhancement against extreme weather that mainly impacts the grid network facilities.

This paper presents and illustrates the methodology to estimate the outage costs originating from transmission component failures using data obtained from an outage cost survey conducted in the year 2017 in the Canadian province of Saskatchewan. Since past outage cost surveys were all focused on outages due to a lack of generation supply, this paper illustrates approximations required due to the lack of data needed to calculate outage costs originating from transmission failures. This paper shows that the amount of customer data required increases with the number of periods in the SPD model for the customer sector. The industrial sector SPD model has only one period, and therefore, the data from other existing surveys can be adequately utilized. The industrial and residential sectors account for over 80% of the total system load in most power systems, and therefore, concentrated efforts are required to plan appropriate survey questionnaires for the residential sector in order to obtain useful data to assess the transmission-originated outage costs. New

surveys that incorporate these recommendations will yield realistic outage costs that can then be used in value-based reliability or resiliency investments in emerging technologies and transmission facilities.

Author Contributions: Conceptualization, R.K.; methodology, R.K.; software, S.B.; validation, S.B.; formal analysis, S.B. and R.K.; investigation, S.B.; resources, R.K.; data curation, S.B.; writing—original draft preparation, S.B.; writing—review and editing, R.K.; visualization, R.K.; project administration, R.K.; funding acquisition, R.K. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by Natural Sciences and Engineering Research Council, Grant number 585294-2023.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The data presented in this study are available on request from the corresponding author.

Conflicts of Interest: The authors declare no conflict of interest.

References

1. Billinton, R.; Allan, R.N. *Reliability Evaluation of Power Systems*; Springer: Boston, MA, USA, 1996; ISBN 978-1-4899-1862-8.
2. Billinton, R. Evaluation of Reliability Worth in an Electric Power System. *Reliab. Eng. Syst. Saf.* **1994**, *46*, 15–23. [\[CrossRef\]](#)
3. Billinton, R.; Oteng-Adjei, J. Cost/Benefit Approach to Establish Optimum Adequacy Level for Generating System Planning. *IEE Proc. C Gener. Transm. Distrib.* **1988**, *135*, 81. [\[CrossRef\]](#)
4. Billinton, R.; Goel, L. A Procedure for Estimating the Overall System Worth Associated with Generating Unit Refurbishment. *Can. J. Electr. Comput. Eng.* **1990**, *15*, 140–148. [\[CrossRef\]](#)
5. Billinton, R.; Adzanu, S. Adequacy and Reliability Cost/Worth Implications of Nonutility Generation. *IEE Proc. Gener. Transm. Distrib.* **1996**, *143*, 115. [\[CrossRef\]](#)
6. Sarkar, A. Reliability Evaluation of a Generation-Resource Plan Using Customer-Outage Costs in India. *Energy* **1996**, *21*, 795–803. [\[CrossRef\]](#)
7. Billinton, R.; Pandey, M. Generating Capacity Planning Criteria Determination for Developing Countries—Case Study of Nepal. *IEE Proc. Gener. Transm. Distrib.* **1999**, *146*, 491. [\[CrossRef\]](#)
8. Prada, J.F.; Ilic, M.D. Pricing Reliability: A Probabilistic Approach. In Proceedings of the Large Engineering Systems Conference on Power Engineering, Halifax, NS, Canada, 20–22 June 1999.
9. Gen, B. Reliability and Cost/Worth Evaluation of Generating Systems Utilizing Wind and Solar Energy. Ph.D. Thesis, University of Saskatchewan, Saskatoon, SK, Canada, 2005.
10. Bagen, B.; Billinton, R. Reliability Cost/Worth Associated with Wind Energy and Energy Storage Utilization in Electric Power Systems. In Proceedings of the 10th International Conference on Probabilistic Methods Applied to Power Systems, Rincon, PR, USA, 25–29 May 2008.
11. Limbu, T.R.; Saha, T.K.; McDonald, J.D.F. Value-Based Allocation and Settlement of Reserves in Electricity Markets. *IET Gener. Transm. Distrib.* **2011**, *5*, 489. [\[CrossRef\]](#)
12. Oteng-Adjei, J.; Malori, A.-M.I.; Anto, E.K. Generation System Adequacy Assessment Using Analytical Technique. In Proceedings of the 2020 IEEE PES/IAS PowerAfrica, Nairobi, Kenya, 25–28 August 2020; pp. 1–5.
13. Billinton, R.; Tollefson, G.; Wacker, G. Assessment of Electric Service Reliability Worth. *Int. J. Electr. Power Energy Syst.* **1993**, *15*, 95–100. [\[CrossRef\]](#)
14. Wacker, G.; Wojczynski, E.; Billinton, R. Interruption Cost Methodology and Results—A Canadian Residential Survey. *IEEE Trans. Power Appar. Syst.* **1983**, PAS-102, 3385–3392. [\[CrossRef\]](#)
15. Wojczynski, E.; Billinton, R.; Wacker, G. Interruption Cost Methodology and Results—A Canadian Commercial and Small Industry Survey. *IEEE Trans. Power Appar. Syst.* **1984**, PAS-103, 437–444. [\[CrossRef\]](#)
16. Wacker, G.; Billinton, R. Farm Losses Resulting from Electric Service Interruptions—A Canadian Survey. *IEEE Trans. Power Syst.* **1989**, *4*, 472–478. [\[CrossRef\]](#)
17. Tollefson, G.; Billinton, R.; Wacker, G.; Chan, E.; Awewa, J. A Canadian Customer Survey to Assess Power System Reliability Worth. *IEEE Trans. Power Syst.* **1994**, *9*, 443–450. [\[CrossRef\]](#)
18. Lawton, L.; Sullivan, M.; Van Liere, K.; Katz, A.; Eto, J. *A Framework and Review of Customer Outage Costs: Integration and Analysis of Electric Utility Outage Cost Surveys*; Lawrence Berkeley National Laboratory: Berkeley, CA, USA, 2003.
19. Sullivan, M.J.; Mercurio, M.; Schellenberg, J. *Estimated Value of Service Reliability for Electric Utility Customers in the United States*; Lawrence Berkeley National Laboratory: Berkeley, CA, USA, 2009; p. LBNL-2132E, 963320.

20. Sanghvi, A.P. Economic Costs of Electricity Supply Interruptions: US and Foreign Experience. *Energy Econ.* **1982**, *4*, 180–198. [CrossRef]
21. Wacker, G.; Tollefson, G. Electric Power System Customer Interruption Cost Assessment. *Reliab. Eng. Syst. Saf.* **1994**, *46*, 75–81. [CrossRef]
22. Outage Cost Estimation Guidebook. Available online: <https://www.epri.com/research/products/TR-106082> (accessed on 11 December 2022).
23. TPL-001-5; Transmission System Planning Performance Requirements. North American Electric Reliability Corporation (NERC): Washington, DC, USA.
24. Papic, M.; Logan, D. Bibliography on Composite System Reliability Assessment 2000–2020. In Proceedings of the 2022 17th International Conference on Probabilistic Methods Applied to Power Systems (PMAPS), Manchester, UK, 12–15 June 2022; pp. 1–8.
25. Billion-Dollar Weather and Climate Disasters | National Centers for Environmental Information (NCEI). Available online: <https://www.ncei.noaa.gov/access/billions/> (accessed on 20 May 2023).
26. Bhattacharai, S.; Sapkota, A.; Karki, R. Analyzing Investment Strategies for Power System Resilience. In Proceedings of the 2022 IEEE Power & Energy Society General Meeting (PESGM), Denver, CO, USA, 17–21 July 2022; pp. 1–5.
27. Gasser, P.; Lustenberger, P.; Cinelli, M.; Kim, W.; Spada, M.; Burgherr, P.; Hirschberg, S.; Stojadinovic, B.; Sun, T.Y. A Review on Resilience Assessment of Energy Systems. *Sustain. Resilient Infrastruct.* **2021**, *6*, 273–299. [CrossRef]
28. Lin, Y. A Review of Key Strategies in Realizing Power System Resilience. *Glob. Energy Interconnect.* **2018**, *1*, 9.
29. Wenyuan, L.; Billinton, R. A Minimum Cost Assessment Method for Composite Generation and Transmission System Expansion Planning. *IEEE Trans. Power Syst.* **1993**, *8*, 628–635. [CrossRef]
30. Siddiqi, S.N.; Baughman, M.L. Value-Based Transmission Planning and the Effects of Network Models. *IEEE Trans. Power Syst.* **1995**, *10*, 1835–1842. [CrossRef]
31. Choi, J.; Mount, T.D.; Thomas, R.J.; Billinton, R. Probabilistic Reliability Criterion for Planning Transmission System Expansions. *IEE Proc. Gener. Transm. Distrib.* **2006**, *153*, 719. [CrossRef]
32. Choi, J.S.; Tran, T.T.; Kang, S.R.; Jeon, D.H.; Lee, C.H.; Billinton, R. A Study on the Optimal Reliability Criteria Decision for a Transmission System Expansion Planning. In Proceedings of the IEEE Power Engineering Society General Meeting, 2004, Denver, CO, USA, 6–10 June 2004; Volume 1, pp. 607–613.
33. Chowdhury, A.A.; Koval, D.O. Reliability Cost-Benefit Evaluation of Transmission Capital Projects. In Proceedings of the 2006 IEEE Power Engineering Society General Meeting, Montreal, QC, Canada, 18–22 June 2006; pp. 607–613.
34. Billinton, R.; Wangdee, W. Reliability-Based Transmission Reinforcement Planning Associated With Large-Scale Wind Farms. *IEEE Trans. Power Syst.* **2007**, *22*, 34–41. [CrossRef]
35. Oliveira, G.C.; Binato, S.; Pereira, M.V.F. Value-Based Transmission Expansion Planning of Hydrothermal Systems Under Uncertainty. *IEEE Trans. Power Syst.* **2007**, *22*, 1429–1435. [CrossRef]
36. Leite Da Silva, A.M.; Rezende, L.S.; Da Fonseca Manso, L.A.; De Resende, L.C. Reliability Worth Applied to Transmission Expansion Planning Based on Ant Colony System. *Int. J. Electr. Power Energy Syst.* **2010**, *32*, 1077–1084. [CrossRef]
37. Sullivan, M.; Schellenberg, J.; Blundell, M. *Updated Value of Service Reliability Estimates for Electric Utility Customers in the United States*; Lawrence Berkeley National Laboratory: Berkeley, CA, USA, 2015; p. LBNL-6941E, 1172643.
38. Dalton, J.G.; Garrison, D.L.; Fallon, C.M. Value-Based Reliability Transmission Planning. *IEEE Trans. Power Syst.* **1996**, *11*, 1400–1408. [CrossRef]
39. Brandenburg, L.; Clarkson, G.; Leo, J.; Asbury, J.; Brandon-Brown, F.; Derderian, H.; Mueller, R.; Swaroop, R. *Load Research Manual. Volume 1. Load Research Procedures*; Argonne National Lab. (ANL): Argonne, IL, USA; National Economic Research Associates: New York, NY, USA, 1980.
40. Puckett, C.; Williamson, C.; Godin, C.; Gifford, W.; Farland, J.; Laing, T.; Hong, T. Utility Load Research: The Future of Load Research Is Now. *IEEE Power Energy Mag.* **2020**, *18*, 61–70. [CrossRef]
41. Billinton, R.; Zhang, W. Cost Related Reliability Evaluation of Bulk Power Systems. *Int. J. Electr. Power Energy Syst.* **2001**, *23*, 99–112. [CrossRef]
42. Wacker, G.; Billinton, R. Customer Cost of Electric Service Interruptions. *Proc. IEEE* **1989**, *77*, 919–930. [CrossRef]
43. Karki, N.R.; Karki, R. *Outage Cost Assessment for Value Based Reliability Investment in System Generating Capacity*; Technical Report Prepared for Saskatchewan Power Corporation; University of Saskatchewan: Saskatoon, SK, Canada, 2018.
44. Inflation Calculator. Available online: <https://www.bankofcanada.ca/rates/related/inflation-calculator/> (accessed on 19 August 2023).

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.